EXNO: 4	**Strategize impedance matching with characteristic impedance and load impedance through the measurement of** **a) Location of the stub** **b) Length of the stub**
DATE:	

Test Case 1:

Optimize impedance matching using single stub matching by finding the location and length of the stub for the values of load impedance of 100 $\underline{30}^{o}$ to 350 $\underline{30}^{o}$ keeping the characteristic impedance Zo= 60 Ω constant .

AIM: To optimize impedance matching using single stub matching by finding the location and length of the stub for the values of load impedance of 100 30° to 35$\underline{0}$ 30° keeping the characteristic impedance Zo= 60 Ω constant .

SOFTWAREREQUIRED:

SCILAB

PROCEDURE:

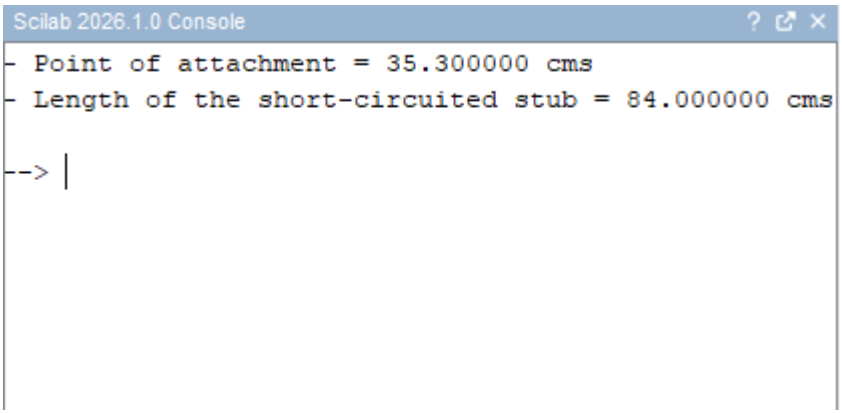
- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

PROGRAM:

```
clear; clear; clc;
ZR=100;Zo=600;f=100*(10^6);
lo=300/(f*(10^-6)); //lo=wavelength Ls=(lo/(2*%pi))*(atan(sqrt(ZR/Zo)));
printf("-Point of attachment = %f cms\n",round(Ls*(10^2)*10)/10)
Lt=(lo/(2*%pi))*(%pi+(atan((sqrt(ZR*Zo))/(ZR-Zo))));
printf("-Length of the short circuited stub = %f cms",round(Lt*(10^2)));  ‘
```

OUTPUT: Variables Z_R; Constants Zo = 60Ω; f=100*(10^6);

Input	Output	
ZR Ohms. angle 30°	**Point of attachment**	**Length of short-circuited stub (cm)**
100	**18.500000 cms**	**128.000000 cms**
200	**25.000000 cms**	**116.000000 cms**
300	**29.400000 cms**	**104.000000 cms**
400	**32.700000 cms**	**94.000000 cms**
500	**35.300000 cms**	**84.000000 cms**



Result :

Impedance matching using single stub matching is optimized by calculating the location and length of the stub f or varying values of load impedance keeping the characteristic impedance Zo= 60 Ω constant .

Test Case 2:

Optimize impedance matching using single stub matching by finding the location and length of the stub for the values of characteristic impedance ranging from 600Ω to 1800 Ω. Keeping the load impedance Z_R= 100 ∠0° constant .

AIM: To optimize impedance matching using single stub matching by finding the location and length of the stub for the values of characteristic impedance ranging from 600Ω to 1800 Ω. Keeping the load impedance Z_R= 100 30° constant .

SOFTWAREREQUIRED:

SCILAB

PROCEDURE:

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

PROGRAM:

```
clear; clear; clc;

ZR=100;Zo=600;f=100*(10^6);

lo=300/(f*(10^-6)); //lo=wavelength

Ls=(lo/(2*%pi))*(atan(sqrt(ZR/Zo)));

printf("-Point of attachment = %f

cms\n",round(Ls*(10^2)*10)/10)

Lt=(lo/(2*%pi))*(%pi+(atan((sqrt(ZR*Zo

))/(ZR-Zo))));

printf("-Length of the short circuited stub = %f cms",round(Lt*(10^2)));
```

OUTPUT:

Variables Z_0;
Constants Z_R= 100 30° Ω; f=100*(10^6);

Input	**Output**	
Zo Ohms.	**Point of attachment**	**Length of short-circuited stub (cm)**
600	**18.500000 cms**	**128.000000 cms**
700	**17.300000 cms**	**130.000000 cms**

800	**16.200000 cms**	**132.000000 cms**
900	**15.400000 cms**	**133.000000 cms**
1000	**14.600000 cms**	**134.000000 cms**

RESULT:

Impedance matching using single stub matching is optimized by calculating the location and length of the stub for varying values of characteristic impedance keeping the load impedance constant .